

LESSON 7-5 • SESSION 1 • GUIDED NOTES

Plotting Points on the Coordinate Plane

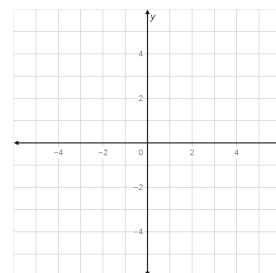
Learning target – I can plot a point with rational coordinates and name the quadrant it lies in. • 6.NS.C.6c

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
coordinate plane <i>plano de coordenadas</i>	A grid made by a horizontal and a vertical _____.	x-axis across, y-axis up and down
ordered pair <i>par ordenado</i>	Two numbers (x, y) that name one _____.	(-3, 4)
origin <i>origen</i>	The point _____ where the two axes cross.	start every plot here
quadrant <i>cuadrante</i>	One of the _____ sections of the plane.	Quadrant II is upper left

KEY IDEA*complete each sentence*

- The x-coordinate tells you to move _____. The y-coordinate tells you to move up or down.
- Signs name the quadrant: $(-, +)$ is Quadrant ____ and $(-, -)$ is Quadrant III.

WORKED EXAMPLE*fill in the blanks as we solve*Plot the point $(-5, -4)$. Name its quadrant.**Step 1** Start at the origin, $(0, 0)$.**Step 2** Move 5 units _____, because $x = -5$.**Step 3** Move 4 units _____, because $y = -4$.**Answer** $(-, -)$ means Quadrant _____, the lower left.**YOU TRY***use the example as a model***1.** Name the quadrant of each point.**a.** $(6, -2)$ _____**b.** $(-1.5, 3)$ _____**2.** Plot and label $A(-4, 2)$ and $B(3, -5)$.**SUM IT UP***in your own words*To plot (x, y) , I start at _____, then I _____.

LESSON 7-5 • SESSION 2 • GUIDED NOTES**Reflecting Points Across Axes**

Learning target – I can reflect a point across the x-axis or the y-axis and name the new coordinates. · 6.NS.C.6b

WORDS TO KNOW*fill in the missing words*

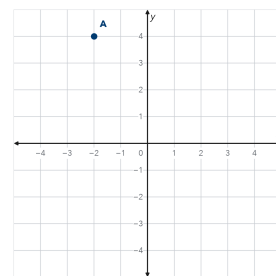
WORD	WHAT IT MEANS	EXAMPLE
reflection <i>reflexión</i>	A ____ that makes a mirror image across a line.	(4, 1) becomes (4, -1) across the x-axis
x-axis <i>eje x</i>	The _____ number line through the origin.	runs left and right
y-axis <i>eje y</i>	The _____ number line through the origin.	runs up and down

KEY IDEA*complete each sentence*

- Across the x-axis, the ____-coordinate changes sign.
- Across the y-axis, the ____-coordinate changes sign.

WORKED EXAMPLE*fill in the blanks as we solve*

Reflect (7, -2) across the x-axis. Then reflect (7, -2) across the y-axis.

Step 1 Across the x-axis: keep $x = 7$ and change -2 to _____. Image: (7, 2).**Step 2** Across the y-axis: keep $y = -2$ and change 7 to ____.**Answer** The images are **(7, 2)** and _____.**YOU TRY***use the example as a model***1.** Write each image.**a.** (3, -5) across the x-axis _____**b.** (-6, 1) across the y-axis _____**2.** Point A is at (-2, 4). Plot A and its reflection B across the x-axis.**B** = _____**SUM IT UP***in your own words*

When I reflect across the y-axis, ____ changes because ____.
